



PROFESSIONAL SUMMARY

Mechanical Engineering candidate bridging physical design, digital engineering, and additive manufacturing workflows. Experienced in establishing baseline technical documentation, performing design validation, and implementing rapid prototyping processes. Adept at converting physical concepts into structured, implementation-ready digital models while collaborating across multidisciplinary, cross-functional teams to optimize engineering communication and product lifecycles.

SELECTED HIGHLIGHTS

- Experienced in translating physical designs into structured engineering documentation and digital manufacturing files.
- Proficient in rapid prototyping and CAD-to-production workflows utilizing major parametric modeling platforms.
- Exposed to multidisciplinary engineering validation pipelines from dimensional inspection through technical reporting.
- Demonstrated ability to facilitate cross-functional collaboration and optimize technical information organization.

TECHNICAL CAPABILITIES

Technical Information & Documentation

- Technical Documentation
- Engineering Specifications
- Technical Reporting
- Drawing Interpretation
- Information Organization

Engineering Analysis & Validation

- Reverse Engineering
- Assembly Validation
- Design Optimization
- Process Analysis
- Tolerance Analysis

Digital Engineering & Systems

- CAD-to-Production Workflows
- Systems Thinking
- Workflow Optimization
- Data Interpretation
- Cross-functional Collaboration

Software & Digital Tools

Fusion 360 Siemens NX SolidWorks AutoCAD Python
MATLAB COMSOL Blender HTML CSS

PROFESSIONAL EXPERIENCE

Reverse Engineering Intern Femlogic Technologies Pvt. Ltd.

June 2026

- Captured high-accuracy dimensional data of mechanical systems using precision metrology tools to establish baseline technical documentation.
- Executed digital engineering workflows to reconstruct physical geometries into parametric 3D models, resolving design deviations and model geometry errors.
- Performed tolerance stack analysis and clearance checks to validate cross-functional compatibility with modern assemblies prior to print layout optimization.

Industrial Engineering Trainee Steel Authority of India Ltd. (SAIL) – Salem Steel Plant

December 2025

- Supported engineering operations through drawing interpretation of heavy steel rolling mills and mechanical system schematics.
- Managed technical documentation by organizing preventative maintenance protocols, operational checklists, and safety manuals.
- Collaborated with maintenance teams on process analysis and data interpretation of equipment logs to trace component wear histories.

Product Design Intern Femlogic Technologies Pvt. Ltd.

June 2025 – July 2025

- Participated in CAD workflows and design validation of structural seating assemblies for high-reliability transit train systems.
- Reviewed structural interface specifications and fasteners selection, ensuring compliance with manufacturing (DFM) requirements and assembly tolerances.
- Drafted technical specifications and conducted tolerance analysis on sheet metal parts to satisfy quality validation reviews.

Production Engineering Intern K.M. Knitwear Pvt. Ltd.

July 2024 – August 2024

- Conducted process analysis on packaging layouts, developing workflow optimization schemes that successfully increased output rates by 10%.
- Established structured shipping layouts and packaging specifications, improving information organization to protect mechanical parts during logistics cycles.
- Provided engineering support during diagnostics and scheduled preventive maintenance, tracing mechanical issues to reduce machinery downtime parameters by 15%.

MAJOR ENGINEERING PROJECTS

Project EcoFloat (Niral Hackathon Regional Winner)

ROS · IoT · Fusion 360 · Arduino · Sensor Fusion

- Co-developed an autonomous robotic catamaran configured to clear floating debris and execute data interpretation of water toxicity levels.
- Implemented a real-time toxicity sensing and mapping framework integrating ROS node communications and IoT data streams, converting field telemetry into structured data sets.
- Executed digital manufacturing workflows in CAD to optimize structural assemblies, sensor mounts, and rapid prototyping of hardware interfaces.

Regenerative Braking System Simulation

MATLAB Simulink · BMS · Powertrain Modeling · Control Engineering

- Simulated an active regenerative braking control loop for EV powertrains in MATLAB Simulink, establishing mathematical models to map brushless motor kinetics and battery recovery rates.
- Executed data interpretation of transient current profiles during deceleration phases to optimize energy recovery, extending simulated battery range parameters by up to 20%.
- Documented system design validation reports detailing energy flow diagrams, safety limits, and thermal stress parameters under variable friction-braking loads.

Advanced Aquatic Monitoring Platform

Hyperspectral Imaging · Sensor Fusion · Optical Gimbals · Additive Manufacturing

- Designed a stabilized floating vessel equipped with a high-resolution optical spectrometer gimbal for real-time remote environmental mapping of lake water quality indicators.
- Processed and analyzed hyperspectral spectral datasets using custom scripts to perform chlorophyll concentration mapping and microplastic polymer detection.
- Optimized structural gimbals and chassis components in CAD, validating designs to meet DfAM and print parameters to minimize wave disturbance interference.

Hemispherical CPV Solar Tracking System

COMSOL Multiphysics · Solar Tracking Math · Fusion 360 · MATLAB

- Designed a Concentrator Photovoltaic (CPV) tracking dome with dual-axis actuators driven by solar vector calculations to maximize daily energy absorption.
- Validated structural integrity, fluid-structure wind loads, and thermal characteristics of the compound panel shell using COMSOL Multiphysics finite element analysis (FEA).
- Created detailed engineering drawings, assembly drawings, and technical documentation specifications detailing clearance stack-ups and mechanical tolerances.

EDUCATION

Bachelor of Mechanical Engineering	St. Joseph's College of Engineering, Chennai, TN	2023 – Present
Higher Secondary Education (Class XII)	Shri Swamy Matriculation Higher Secondary School, Salem, TN	2022 – 2023
Secondary School Education (Class X)	Shri Swamy Matriculation Higher Secondary School, Salem, TN	2020 – 2021

CERTIFICATIONS

Autodesk Certified Fusion Designer Professional CAD Design Credential	Certified Solidworks Designer Mechanical Modeling & Validation Spec
Certified Siemens NX CAD Designer Advanced Parametric Modeling Specialist	AutoCAD for 2D Modelling Technical Drafting & Detailing Standard
Certified in Python Advanced Computational Data Structures & Integration	Certified Frontend Developer Core Interface Architect (HTML/CSS/JS)
Blender for 3D Modelling Complex Surface Mesh Optimization	C++ Programming Object-Oriented Software & Logic Design

ADDITIONAL INFORMATION

Languages English – Fluent Tamil – Native	Professional Strengths Structured Technical Communication · Analytical Thinking · Fast Learning Ability · Attention to Detail · Engineering Collaboration · Digital Tool Adaptability
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